

DisorderFlow

An Auditable Platform for Antibody Design against Disordered Epitopes

Team 28 - Analyzing Disorder

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AI-driven Life Science Discovery | iBEC 2026

1. Why Disordered Epitopes?

Rigid-pose antibody design is poorly matched to conformational heterogeneity

- A β , tau and α -synuclein are central protein-misfolding targets
- One antigen sequence can occupy multiple structural states
- A candidate should not be selected by one pose or one uncalibrated confidence score
- Engineering goal: auditable generation, counterfactual controls and uncertainty-aware filtering

2. Original Contributions

A platform, not a single opaque score

- ECLS: complex versus peptide-stripped structural likelihood contrast
- Position-sensitive antigen-disorder routing in a Bayesian Flow Network
- BFN expands H3 exploration, with an explicitly measured developability tradeoff
- Fail-closed binder confidence: abstain when experimental labels are absent

3. End-to-End Platform

Structure → generation → multi-state scoring → filtering → validation → handoff

- BFN, ProteinMPNN and ESM-IF matched workflows
- Gradio interface plus reproducible command-line runners
- AF2-Multimer through a persistent WSL2 GPU worker
- Developability, diversity, provenance and claim-boundary exports

4. Frozen ECLS Temporal Result

31 structures | 15 antigen clusters | 200 shuffles per record

- Mean native advantage: 0.1723
- Cluster-bootstrap 95% CI: [0.0592, 0.2919]
- Positive clusters: 12/15 (80%)
- Bounded claim: structural sequence contrast, not affinity

5. Aβ Candidate Engineering Funnel

A frozen, auditable 3D6/4HIX case study

- 481 source memberships → 352 unique hypotheses
- 77 pass computational gates → 24 diverse candidates
- 27 entities × 3 AF2 seeds = 81/81 successful predictions
- 16 pass AF2/PRODIGY gates → 12 final computational candidates

6. Multi-Seed Structural Validation

Small score changes are reported with uncertainty

- Native median ipTM: 0.4762
- Candidate median ipTM range: 0.4763-0.5034
- Candidate seed ranges: 0.0030-0.0087
- Interface PAE $\approx$ 26: candidates remain computational hypotheses

7. Evidence Audit and Second Target

Correcting claims is part of the engineering workflow

- 1,289 records × 5 conformations; 219 have exact DisProt disorder-region evidence
- The unsupported phrase '800+ natural IDPs' has been removed
- Tau/5MP3: correct 13-aa H3, five poses, 275 unique hypotheses, 163 pass all gates
- Eight diverse Tau candidates; minimum pairwise Hamming distance 2

8. Engineering and Translation

Computational output is converted into an auditable handoff

- Secreted scFv contract: signal-VH-(G4S)3-VL-G4S-His6
- 12 candidates plus native and counterfactual sequence controls
- Blinded assay layouts and measurement import schemas
- No false probability: binder-confidence status is abstain_not_trained

9. iBEC Deliverables

Primary track: AI-driven Life Science Discovery

- Runnable platform and 4HIX demonstration workflow
- Frozen ECLS benchmark and 449-artifact release lineage
- A β 352 $\rightarrow$ 24 $\rightarrow$ 12 case study with 81 hashed AF2 structures
- Completed audits and Tau 275 $\rightarrow$ 163 $\rightarrow$ 8 second-target extension
- Current boundary: no experimental binding, affinity or therapeutic claim